

Homework 4

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1. Recall that a metric space (X, d) is called complete if every Cauchy sequence in X converges. Show that if X is compact then it must be complete.

Proof. Let (a_n) be a Cauchy sequence in X . Since X is a compact metric space, it is sequentially compact and thus (a_n) has a subsequence (b_n) which converges to a point $p \in X$. Since (a_n) is Cauchy, it converges to the same $p \in X$, and thus X is complete. $\square$

2. Consider $C([0, 1])$ with the C^0 metric. Let

$$B = \{f \in C([0, 1]) : d(f, \mathbf{0}) \leq 1\}$$

where $\mathbf{0}$ is the constant function $g \equiv 0$. Note that B is closed and bounded. Show that it is not compact.

Proof. Define

$$f_n(x) = \begin{cases} 2nx & \text{if } x \in [0, 1/2n) \\ 2 - 2nx & \text{if } x \in [1/2n, 1/n) \\ 0 & \text{otherwise.} \end{cases}$$

Then each $f_n(x) \in C([0, 1])$, and the uniform limit of the sequence as $n \rightarrow \infty$ is

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & \text{otherwise.} \end{cases}$$

Thus any subsequence must also converge to f , but of course f is not continuous on $[0, 1]$ since it has a jump discontinuity at $x = 0$, so B is not sequentially compact. $\square$

3. Consider $C([0, 1])$ with the L^1 metric. Let

$$B = \{f \in C([0, 1]) : d(f, \mathbf{0}) \leq 1\}$$

and show that it is not compact.

Proof. Define

$$f_n(x) = \begin{cases} 4n^2x & \text{if } x \in [0, 1/2n) \\ 4n - 4n^2x & \text{if } x \in [1/2n, 1/n) \\ 0 & \text{otherwise.} \end{cases}$$

Then each $f_n \in C([0, 1])$, and the integral of each f_n over $[0, 1]$ is 1, so each $f_n \in B$. However, since these do not converge, there is no convergent subsequence, and thus X is not compact. $\square$

4. Let $K_1 \supset K_2 \supset K_3 \supset \cdots$ be a sequence of non-empty compact subsets of a metric space (X, d) . Show that $\bigcap_{n=1}^{\infty} K_n$ is also non-empty.

Proof. For each $n > 0$, let $a_n \in K_n$. Then $(a_n) \subset K_1$, so (a_n) has a subsequence (a_{i_m}) which converges to a point $p \in K_1$ by sequential compactness of K_1 . However, for each $n > 0$, $(a_m)_{m=n}^{\infty} \subset K_n$, so indeed all but finitely many entries of (a_{i_m}) also lie in K_n , and thus their limit (which is unique amongst the subsequential limits of (a_{i_m})) is in K_n by compactness of K_n . Since this is true for all n , we get that $p \in K_n$ for all n , and thus $p \in \bigcap K_n$. $\square$